

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XIV

JEE (Main)-2022

TEST DATE: 09-06-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

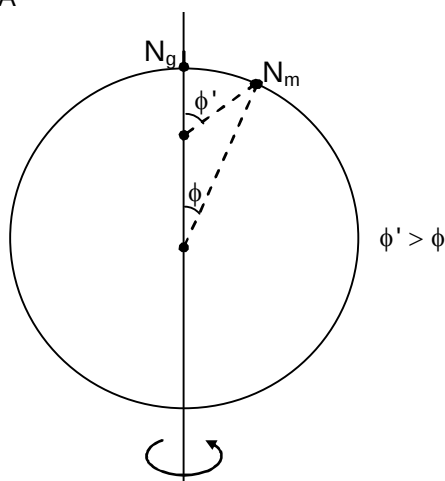
SECTION – A

1. D

Sol. Net torque about center of mass is zero. Therefore, sphere will not rotate.

2. A

Sol.

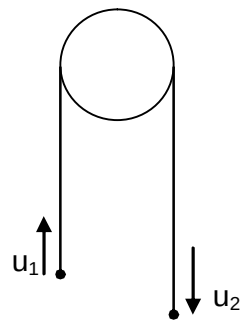


3. C

 Sol. $u_1 = u_2$

$$\omega_1 R - v_1 = v_2 - \omega_2 R$$

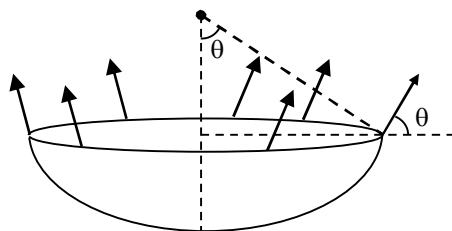
$$v_1 + v_2 = (\omega_1 + \omega_2) R$$



4. B

 Sol. $S \times 2\pi r \sin \theta = mg$

$$S \times 2\pi r \times \frac{r}{R} = mg \quad \dots(i)$$



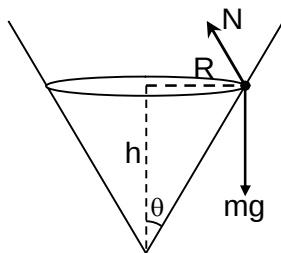
$$R^2 = r^2 + (R - y)^2$$

$$\Rightarrow R = \frac{r^2}{2y}$$

$$\therefore \frac{S \times 2\pi r^2}{r^2} \times 2y = mg$$

$$\Rightarrow S = \frac{mg}{4\pi y}$$

5. C

 Sol. $N \sin \theta = mg$


$$N \cos \theta = m \omega^2 R = m \omega^2 h \tan \theta$$

$$\therefore \tan \theta = \frac{g}{\omega^2 h \tan \theta}$$

$$\omega \propto \frac{1}{\sqrt{h}}$$

$$T \propto \sqrt{h}$$

$$\therefore (C)$$

6. D

Sol. $V_p = -v \times \text{slope}$
 $\therefore$ (D)

7. C

Sol. $B = \frac{\mu_0 i}{4\pi d} \{ \sin(\theta - 90^\circ) + \sin 90^\circ \} + 0$
 $= \frac{\mu_0 i}{4\pi d} (1 - \cos \theta)$

8. C

Sol. $V_{rms} \propto \sqrt{T} \Rightarrow \sqrt{\frac{T_2}{T_1}} = 1.5$
 $T_1 V_1^{\frac{5}{3}-1} = T_2 V_2^{\frac{5}{3}-1}$
 $\frac{V_1}{V_2} = \left(\frac{3}{2}\right)^3$

9. D

Sol. $\frac{V}{E} = \frac{kQ}{\sqrt{R^2 + r^2}} \times \frac{(R^2 + r^2)^{\frac{3}{2}}}{kQ r} = 10 \Rightarrow r = 1, 9$
 $\therefore$ Separation = 8m.

10. A

Sol. $h \propto \frac{1}{g}$
 $\Rightarrow \frac{h_1}{h_2} = \frac{g_2}{g_1} = \frac{g\left(1 - \frac{d}{R}\right)}{g} = 1 - \frac{d}{R}$

11. A

Sol. Resistance of depletion region (no free charge carriers) is very high as compared to resistance of p and n side

12. D

Sol. p, d & v are not independent of each other.

13. B

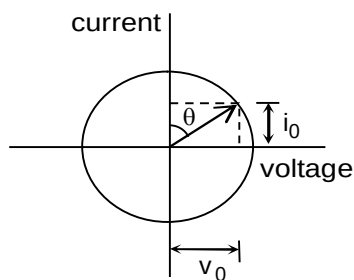
Sol. All photoelectrons do not reach anode at given potential.

14. C

Sol. Each point on the disc oscillates on a circular arc of radius l.

15. B

Sol. $i = i_0 \cos \theta = i_0 \sin(\omega t + \frac{\pi}{2})$
 $v = v_0 \sin \theta = v_0 \sin \omega t$
 so current leads the voltage by 90°



16.

A

Sol.

Mirror is convex because image of a real object is small, erect and virtual. As the object pin moves towards the mirror size of image increase but m is always $m \leq 1$, $m = 1$ when object is at pole]

17.

B

Sol.

$$(\Delta E)_A > (\Delta E)_B$$

$$E_A < E_B$$

$$K.E._A > K.E._B$$

$$\text{So, } U_A > U_B$$

18.

B

Sol.

$$I = I_0 \left(\frac{R}{R + R_G} \right) = 0.5A$$

19.

A

Sol.

$$(\mu - 1)t = n\beta$$

$$\frac{(\mu_1 - 1) \times 1.8 \times 10^{-5}}{(\mu_2 - 1) \times 3.6 \times 10^{-5}} = \frac{18\beta}{9\beta}$$

$$(\mu_1 - 1) = 4(\mu_2 - 1)$$

$$4\mu_2 - \mu_1 = 3$$

20.

C

Sol.

Higher is the binding energy per nucleon more is the stability of nucleus.

$${}^4_2\text{He} \Rightarrow \frac{28}{4} = 7 \text{ MeV/Nucleon}$$

$${}^7_3\text{Li} \Rightarrow \frac{52}{7} = 7.42 \text{ MeV/Nucleon}$$

$${}^{12}_6\text{C} \Rightarrow \frac{90}{12} = 7.5 \text{ MeV/Nucleon} - \text{Most stable}$$

$${}^{14}_7\text{N} \Rightarrow \frac{98}{14} = 7 \text{ MeV/Nucleon}$$

SECTION – B

21. 00001.50

Sol. $d_1 = ut - \frac{1}{2}gt^2 = 40 \times 6 - \frac{1}{2} \times 5 \times 6^2 = 150\text{m}$

$$d_2 = \frac{4^2}{2g} + \frac{1}{2}g\left(t - \frac{T}{2}\right)^2 = \frac{40^2}{2 \times 10} + \frac{1}{2} \times 10 \left(6 - \frac{8}{2}\right)^2$$

$$= 100\text{ m}$$

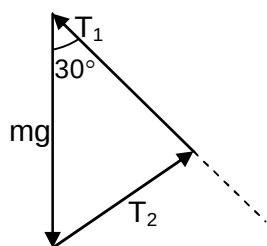
$$\therefore \frac{d_1}{d_2} = \frac{3}{2} = 1.5$$

22. 00060.00

Sol. T_2 will be minimum when

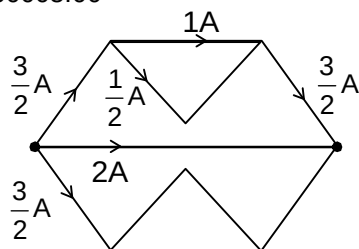
$$T_1 \perp T_2$$

$$\therefore \theta = 60^\circ$$



23. 00005.00

Sol.



$$i = 5A$$

24. 00346.41

Sol. $W_1 = (mg \sin \theta_1 + \mu mg \cos \theta_1)l$

$$400 = 40 \left(\frac{1}{2} + \mu \times \frac{\sqrt{3}}{2} \right) \times 5$$

$$\mu = \sqrt{3}$$

$$W_2 = mg(\sin \theta_2 + \mu \cos \theta_2)l$$

$$= 40 \left(\frac{\sqrt{3}}{2} + \sqrt{3} \times \frac{1}{2} \right) \times 5$$

$$= 200\sqrt{3}\text{J} = 346.41$$

25. 00007.60

Sol. $I = \frac{M}{2}(R_1^2 + R_2^2)$

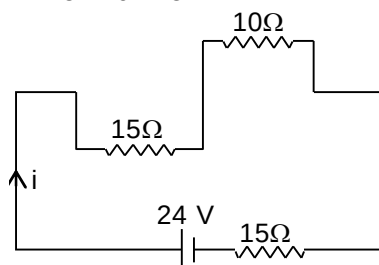
$$dI = \frac{M}{2}(2R_1 dR_1 + 2R_2 dR_2) + \frac{dM}{2}(R_1^2 + R_2^2)$$

$$= 47.5$$

$$\frac{dI}{I} \times 100 = 7.6\%$$

26. 00382.50

Sol. $i = \frac{24}{15 + 10 + 15} = 0.6 \text{ A}$



$$U = \frac{1}{2} \times 5 \times 10^{-6} \times (15 \times 0.6)^2 + \frac{1}{2} \times 10 \times 10^{-6} (10 \times 0.6)^2$$

$$= 382.5 \mu\text{J}$$

27. 00000.50

Sol. $g \sin \theta - \mu g \cos \theta = \frac{g \sin \theta}{1 + 1}$

$$\mu = \frac{\tan \theta}{2} = 0.5$$

28. 00004.00

Sol. $T = 4 \sqrt{\frac{2h}{g}} = 4 \text{ s}$

29. 00020.00

Sol. $C = \frac{q}{V} \Rightarrow C_1 = 6 \mu\text{F}, C_2 = 4 \mu\text{F}, C_3 = 2 \mu\text{F}$

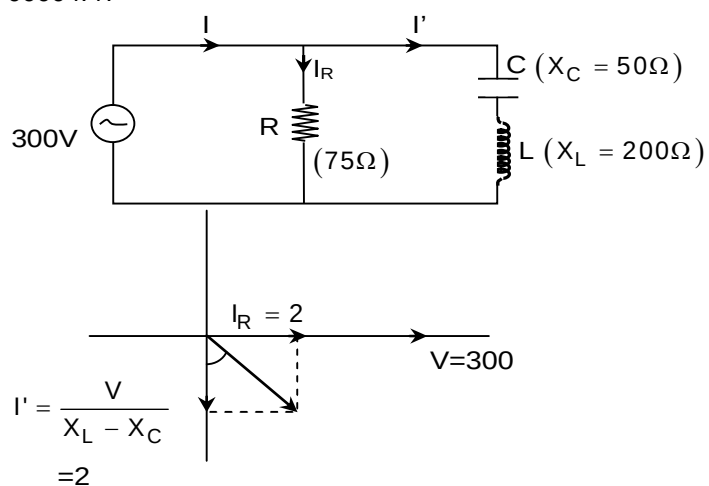
$$\Rightarrow \text{if } V = 10, C_{\text{eq}} = 3 \mu\text{F}$$

$$Q_{\text{total}} = 30 \mu\text{C}$$

$$Q_2 = 30 \times \frac{4}{6} \mu\text{C}$$

30. 00004.47

Sol.

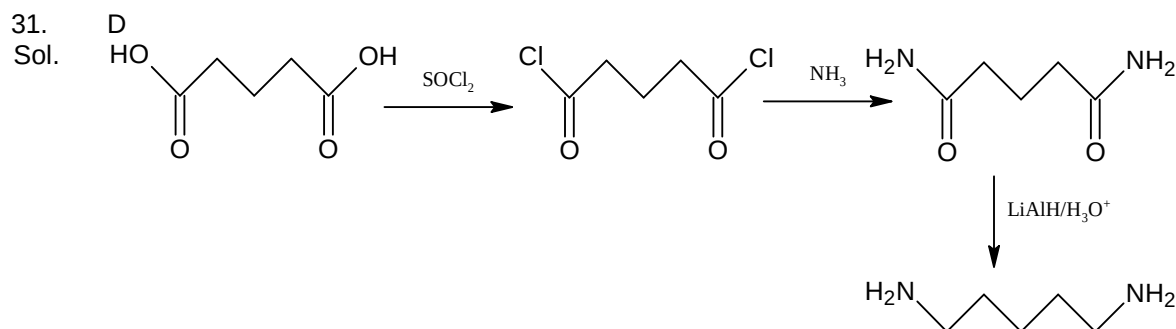


$$I_R = \frac{V}{R} = \frac{300}{75} = 4A, \quad I' = \frac{V}{X_L - X_C} = \frac{300}{150} = 2A, \quad I = \sqrt{I_R^2 + I'^2} = \sqrt{20} = 4.47 \text{ Amp.}$$

Chemistry

PART – B

SECTION – A



32. B
Sol. SeO_2 convert $-\text{CH}_2$ group into $>\text{C}=\text{O}$, adjacent to $>\text{C}=\text{O}$.

33. C
Sol. Phenol is soluble in NaOH, alcohol are also soluble, but not more than 6C. It give only one monobromo, It mean other group present at p-.

34. B
Sol. Argentite is sulphide.

35. A
Sol. ZnSO_4 is soluble in excess of NH_4OH .

$$\text{ZnSO}_4 + 4\text{NH}_4\text{OH} \longrightarrow [\text{Zn}(\text{NH}_3)_4]\text{SO}_4 + 4\text{H}_2\text{O}$$

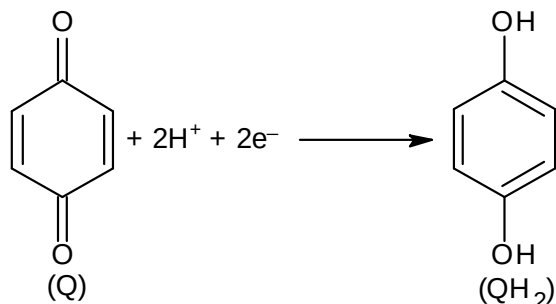
36. D
Sol. $\text{HgCl}_2 + 2\text{KI} \longrightarrow \text{HgI}_2 + 2\text{KCl}$
 $\text{HgI}_2 + 2\text{KI} \longrightarrow \text{K}_2[\text{HgI}_4]$

$$2\text{K}_2[\text{HgI}_4] + 3\text{KOH} + \text{NH}_4\text{Cl} \longrightarrow \begin{array}{c} \text{Hg}-\text{NH}_2 \\ | \\ \text{O} \\ | \\ \text{Hg}-\text{I} \end{array}$$
 (Brown)

37. B
Sol. $\text{PbO}_2 + \text{HNO}_3 + \text{MnSO}_4 \longrightarrow \text{HMnO}_4$

38. D
Sol. In $\text{K}_4[\text{Fe}(\text{CN})_6]$, iron is present in +2 oxidation state.

39. A
Sol.



$$E_{\text{Q/QH}_2} = E_{\text{Q/QH}_2}^\circ - \frac{.0591}{2} \log \frac{1}{[\text{H}^+]^2}$$

$$\text{or } E_{\text{Q/QH}_2} = E_{\text{Q/QH}_2}^\circ - \frac{.0591}{1} \log \frac{1}{[\text{H}^+]}$$

$$\begin{aligned} 0.492 &= 0.699 - \frac{.0591}{1} P_{\text{H}} \\ .0591 P_{\text{H}} &= 0.699 - 0.492 \\ P_{\text{H}} &= 3.5 \end{aligned}$$

40. C

Sol. d_{xy} orbital have probability of finding e^- in XY-plane and in the same plane perpendicular to it.

41. A

Sol.
$$\frac{v_1}{t_1} \times \frac{t_2}{v_2} = \frac{\Delta P_1}{\Delta P_2} \sqrt{\frac{T_2}{T_1}}$$

$$\frac{1}{5} \times \frac{t_2}{1} = \frac{1\sqrt{546}}{1\sqrt{273}}$$

$$t_2 \approx 7.1 \text{ min}$$

42. A

Sol. $\pi = iCRT$

$$\pi = 1.9 \times \frac{0.85}{85 \times 0.1} \times 0.0821 \times 300 \approx 4.67 \text{ atm}$$

43. D

Sol. In Frenkel defect density of crystal does not change.

44. C

Sol. $P_{\text{OH}} = -\log[\text{OH}^-] = -\log \sqrt{K_b C}$

$$\text{Antilog}(-3.92) = \sqrt{K_b C}$$

$$(1.2 \times 10^{-4}) = \sqrt{K_b C} \text{ and } C = \frac{1.7}{303}$$

$$\frac{1.44 \times 10^{-8}}{1.7} \times 303 = K_b$$

$$K_b = 2.62 \times 10^{-6}$$

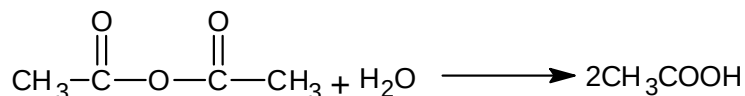
45. A

 Sol. For a first order reaction, $t_{1/2}$ remains constant, if any one reactant taken in excess, then the order with respect to that is zero.

$$\left[\frac{dx}{dt} = k[\text{NO}]^0 [\text{H}_2]^1 \right]$$

46. C

Sol.


 69 gm acetic acid associated with 6 gm H_2O , 18 gm water react with 102 gm $(\text{CH}_3\text{CO})_2\text{O}$

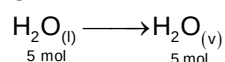
 So, 34 gm acetic anhydride consume 6 gm H_2O .

47. A

Sol. According to Le Chatelier's principle, On increasing pressure equilibrium favour in that direction having lesser volume and for a endotherm reaction, on increasing temperature equilibrium favour in forward direction.

48. C

Sol.



$$\Delta H = \Delta E + \Delta n g R T$$

$$9.72 \times 5 = \Delta E + 5 \times 2 \times 10^{-3} \times 373$$

$$\Delta E = 48.6 - 3.73$$

$$= 44.87 \text{ KCal}$$

49. C

Sol. Freundlich's isotherm related to multimolecular layer, but it fails at very high pressure. Langmuir isotherm explain monolayer adsorption.

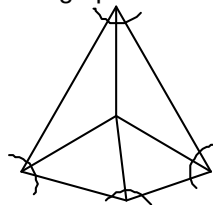
50. C

 Sol. $\text{CCl}_3\text{COO}^\ominus$ on heating generate: CCl_2 (dichlorocarbene). This reaction is related to insertion of carbene in double bond.

SECTION – B

51. 00018.00

Sol. 6 edge present in a tetrahedron. If we cut four corner, 12 edge new created.



52. 00013.00

Sol.

$$X_{\frac{6}{8+3}} Y_6$$

$$X_{\frac{30}{8}} Y_6$$

$$X_{30} Y_{48}$$

$$\text{or } X_5 Y_8$$

53. 00075.00

Sol. $i = \frac{\text{Theoretical M.W}}{\text{Observed M.W}}$

$$1 + 2\alpha = \frac{164}{65.6}$$

$$2\alpha = 2.5 - 1$$

$$\alpha = \frac{1.5}{2} = .75$$

% degree of dissociation = 75%

54. 00075.00



On solving

$$s = 1.5 \times 10^{-7}$$

$$s_1 = 0.5 \times 10^{-7}$$

Simultaneous solubility = $(s + s_1)$

$$= 2 \times 10^{-7}$$

$$\Lambda_m^{\circ} = \frac{k \times 1000}{C}$$

$$\Lambda_m^{\circ} = \frac{290 \times 10^{-10} \times 1000}{2 \times 10^{-7}}$$

$$\Lambda_m^{\circ} = 145 \text{ Scm}^2 \text{ mol}^{-1} = I_{(P^+)}^{m(0)} + I_{(Q^-)}^{m(0)} + I_{(R^-)}^{m(0)}$$

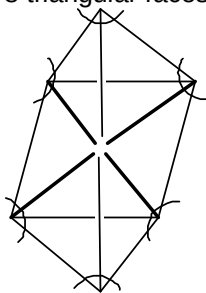
$$145 = 50 + X + 20$$

$$X = 145 - 70$$

$$X = 75 \text{ Scm}^2 \text{ mol}^{-1}$$

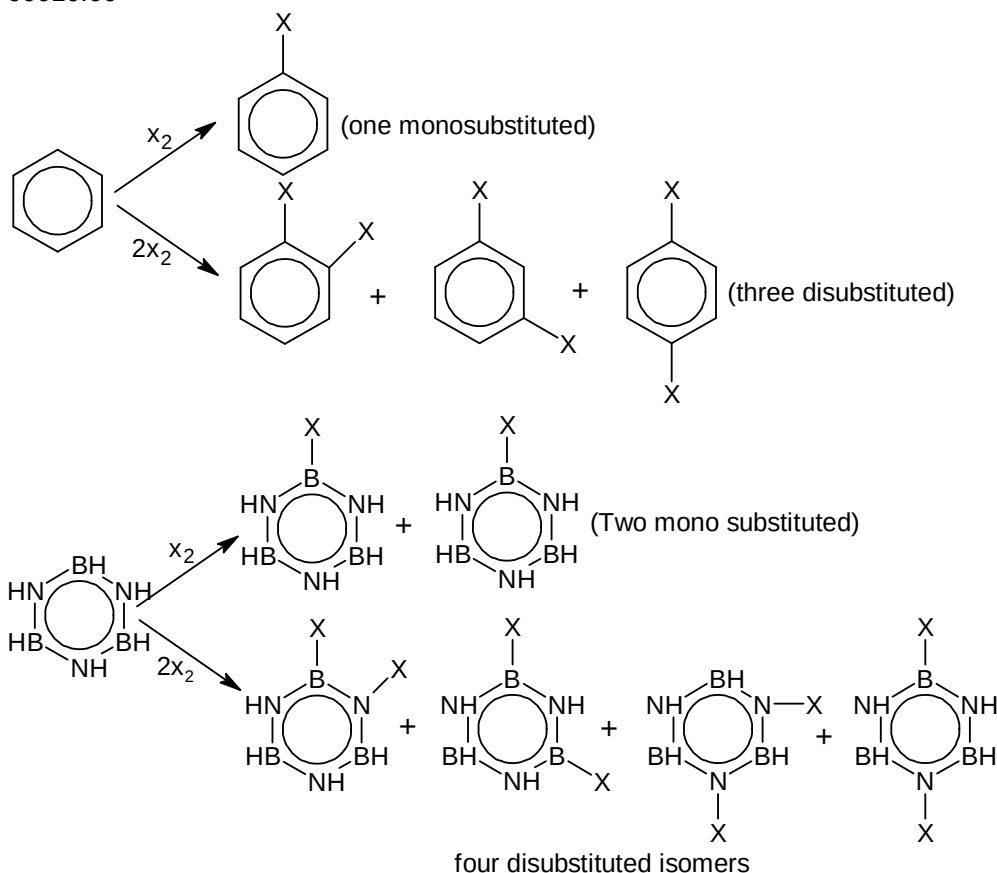
55. 00014.00

Sol. 8 triangular faces gives 8 hexagonal faces and cut of a corner give a cubical faces.



56. 00010.00

Sol.



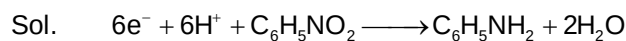
57. 00024.10

Sol.

$$\frac{\ln 2}{t_{1/2}} = \frac{1}{10} \ln \frac{.04}{.03}$$

$$t_{1/2} = 24.10$$

58. 00030.00



59. 00080.00

Sol. Mass of one atom = $\frac{\text{Atomic mass}}{\text{Avogadro no.}}$

$$3.32 \times 10^{-23} = \frac{\text{Atomic mass}}{6.02 \times 10^{23}}$$

Atomic mass ≈ 20

60. 00016.00

Sol.

$$AB_{(g)} \rightleftharpoons A_{(g)} + B_{(g)}$$

$$\frac{1}{1-x} = \frac{x}{x} + \frac{x}{x}$$

$$K_p = 2.56 \times 10^{-2} = \frac{x^2}{1-x^2}$$

$$x = 0.16$$

% dissociation = 16%

Mathematics**PART – C****SECTION – A**

61. C

Sol. $(x-2)^2 + (y-3)^2 + (z-4)^2 = r^2$

Put (0, 1, 0)

$$r^2 = 24$$

62. D

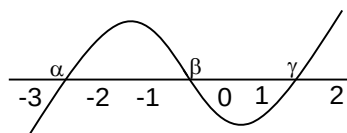
Sol. $\sin^{-1} \frac{|2 \times 3 + (-1) \times 6 + 2 \times (-2)|}{\sqrt{2^2 + (-1)^2 + 2^2} \cdot \sqrt{3^2 + 6^2 + (-2)^2}}$

63. A

Sol. $\frac{N}{x_1 + x_2 + x_3 + x_4} = \frac{1000x_1 + 100x_2 + 10x_3 + x_4}{x_1 + x_2 + x_3 + x_4}$
 $= 1000 - \frac{(900x_2 + 990x_3 + 999x_4)}{x_1 + x_2 + x_3 + x_4} \Rightarrow \text{Maximum value of } \frac{N}{x_1 + x_2 + x_3 + x_4} = 1000$

64. D

Sol.



$$f(0) = -1, f(x) = -4, f(2) = 1, f(-1) = 4, f(-2) = 5, f(-3) = -4$$

$$[\alpha] + [\beta] + [\gamma] = -3 - 1 + 1 = -3$$

65. D

Sol.

$$\left| z - \frac{4}{z} \right| = 2,$$

$$\left| z \right| - \frac{4}{\left| z \right|} \leq \left| z - \frac{4}{z} \right| = 2$$

$$\Rightarrow \left| z \right|^2 - 4 \leq 2\left| z \right|$$

$$\Rightarrow \left| z \right|^2 - 2\left| z \right| - 4 \leq 2$$

$$\Rightarrow \left| z \right| \leq 1 + \sqrt{5}$$

$$\Rightarrow \text{max of } \left| z \right| \text{ is } 1 + \sqrt{5}$$

66. D

Sol.

$$x_k \geq 0, k = 0, 1, 2, 3, 4, 5$$

and

$$x_1 + x_2 + x_3 + x_4 + x_5 = 20 \text{ and } x_1 + x_2 + x_3 = 5$$

$$\Rightarrow x_4 + x_5 = 20 - 5$$

$$\Rightarrow x_4 + x_5 = 15$$

No. of integral solution

$$\binom{15+2-1}{2-1} = \binom{16}{1} = 16 \text{ and number of integral solution of } x_1 + x_2 + x_3 = 5 \text{ is}$$

$$\binom{5+3-1}{3-1} = \frac{7}{2} = \frac{7 \times 6}{2} = 21$$

Hence total number of integral solution is $21 \times 16 = 336$

67. D

$$\begin{aligned} \text{Sol. } \int \frac{\operatorname{cosec}^2 x - 2005}{\cos^{2005} x} dx &= \int (\cos x)^{-2005} \cdot \operatorname{cosec}^2 x dx - 2005 \int \frac{dx}{\cos^{2005} x} \\ &= (\cos x)^{-2005} \cdot (-\cot x) - \int (-2005)(\cos x)^{-2006} (-\sin x)(\cot x) dx - 2005 \int \frac{dx}{\cos^{2005} x} \\ &= -\frac{\cot x}{(\cos x)^{2005}} + C \end{aligned}$$

68. C

$$\text{Sol. Given } 7a - 9b = 0 \Rightarrow b = \frac{7}{9}a$$

Hence number of pairs (a, b) can be (9, 7); (18, 14); (27, 21); (36, 28).

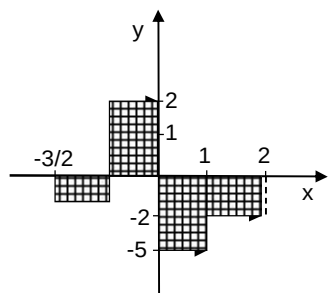
$$\text{Hence required probability} = \frac{4}{{}^{39}C_2} = \frac{4}{741}.$$

69. A

$$\begin{aligned} \text{Sol. } \sum_{r=0}^m \Delta r &= \begin{vmatrix} \sum_{r=0}^m (2r-1) & \sum_{r=0}^m {}^m C_r & \sum_{r=0}^n 1 \\ m^2 - 1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin(m^2) \end{vmatrix} \\ &= \begin{vmatrix} m^2 - 1 & 2^m & m+1 \\ m^2 - 1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin(m^2) \end{vmatrix} \\ &= 0 \end{aligned}$$

70. A

$$\begin{aligned} \text{Sol. } 3[x] - 5 \frac{|x|}{x} &= 3[x] - 5, \text{ if } x > 0 \\ &= 3[x] + 5, \text{ if } x < 0 \end{aligned}$$



$$\begin{aligned}
 &\Rightarrow \int_{-3/2}^2 f(x) dx \\
 &= -1 \left(1 + \frac{3}{2} \right) + 2(1) + 1(-5) + (-2) \\
 &= -\frac{1}{2} + 2 - 5 - 2 = -\frac{11}{2}
 \end{aligned}$$

71. A

Sol. $\cos \theta + \cos 3\theta + \cos 5\theta = \frac{1}{2}$

When $\theta = \pi/7$

72. C

Sol. $\frac{\tan^{-1}(x+h) - \tan^{-1}x}{h} = \sin^2 y - h \sin^2 y \cdot \frac{\sin 2y}{2} + \dots$

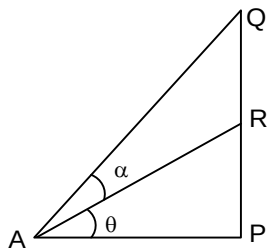
Now, taking limits of both sides as $h \rightarrow 0$, we get

$$\begin{aligned}
 \frac{1}{1+x^2} &= \sin^2 y \Rightarrow x^2 = \operatorname{cosec}^2 y - 1 = \cot^2 y \\
 &\Rightarrow x = \cot y, y = \cot^{-1}(x)
 \end{aligned}$$

73. A

Sol. $\tan(\theta + \alpha) = \frac{PQ}{AP} = \frac{1}{n}$

$$\tan \theta = \frac{PR}{AP} = \frac{(1/2)PQ}{AP} = \frac{1}{2n}$$



$$\alpha = (\theta + \alpha) - \theta$$

$$\Rightarrow \tan \alpha = \frac{\tan(\theta + \alpha) - \tan \theta}{1 + \tan(\theta + \alpha) \tan \theta}$$

$$\frac{\frac{1}{n} - \frac{1}{2n}}{1 + \frac{1}{n} \cdot \frac{1}{2n}} = \frac{n}{2n^2 + 1}$$

74. B

Sol. $f(x) = \begin{vmatrix} x+a^2 & x^4+1 & 3 \\ x+b^2 & 2x^4+2 & 3 \\ x+c^2 & 3x^4+7 & 3 \end{vmatrix}$

$$\Rightarrow f'(x) = \begin{vmatrix} x+a^2 & x^4+1 & 3 \\ x+b^2 & 2x^4+2 & 3 \\ x+c^2 & 3x^4+7 & 3 \end{vmatrix} + \begin{vmatrix} x+a^2 & 4x^3 & 3 \\ x+b^2 & 8x^3 & 3 \\ x+c^2 & 12x^3 & 3 \end{vmatrix} + \begin{vmatrix} x+a^2 & x^4+1 & 0 \\ x+b^2 & 2x^4+2 & 0 \\ x+c^2 & 3x^4+7 & 0 \end{vmatrix} = 0$$

$$f'(x) = 0 + 12x^3 \begin{vmatrix} x+a^2 & 1 & 1 \\ x+b^2 & 2 & 1 \\ x+c^2 & 3 & 1 \end{vmatrix} + 0 = 0$$

75. B

Sol. Given integral $= \int \frac{\cos^2 \theta - \sin^2 \theta}{(\cos \theta + \sin \theta)^2} d\theta$

$$= \int \frac{\cos \theta - \sin \theta}{\sin \theta + \cos \theta} d\theta = \log |\sin \theta + \cos \theta| + c$$

76. A

Sol. $\lim_{x \rightarrow 0} \left[f(x) + \log \left(1 - \frac{1}{e^{f(x)}} \right) - \log(f(x)) \right]$

$$= \lim_{x \rightarrow 0} \left[\log(e^{f(x)}) + \log \left(\frac{1 - \frac{1}{e^{f(x)}}}{f(x)} \right) \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \log \left(\frac{e^{f(x)} - 1}{f(x)} \right) = 0 \Rightarrow \lim_{x \rightarrow 0} \left(\frac{e^{f(x)} - 1}{f(x)} \right) = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) = 0 \Rightarrow f(0) = 0 \text{ (as } f(x) \text{ is a continuous function).}$$

77. B

Sol. $(3)^6 = 729 < 900$ and $(3)^7 = 2187 > 900$

78. C

Sol. Draw ten lines $y = x + r$ and other ten lines $y = -x + r$. We can observe that required squares are $8^2 = 64$

79. D

Sol. We have $x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots \infty = \frac{x^2}{1 - \left(\frac{-x^2}{2} \right)} = \frac{2x^2}{2 + x^2}$

$$x - \frac{x^2}{2} + \frac{x^3}{4} - \dots \infty = \frac{x}{1 - \left(\frac{-x}{2} \right)} = \frac{2x}{2 + x}$$

The given equation becomes

$$\sin^{-1} \left(\frac{2x^2}{2 + x^2} \right) = \frac{\pi}{2} - \cos^{-1} \left(\frac{2x}{2 + x} \right)$$

$$\Rightarrow \frac{2x^2}{2 + x^2} = \frac{2x}{2 + x} \Rightarrow x = 1$$

$$\Rightarrow \cot^{-1}(x) \Rightarrow \cot^{-1}(1) = \frac{\pi}{4}$$

80. C

Sol. Combined mean,

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$= \frac{200 \times 25 + 300 \times 10}{500} = 16$$

$$\text{Let } d_1 = \bar{x}_1 - \bar{x} = 25 - 16 = 9$$

$$\text{and } d_2 = \bar{x}_2 - \bar{x} = 10 - 16 = -6$$

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

$$= \frac{200(9 + 81) + 300(16 + 38)}{500}$$

$$= \frac{33600}{500} = 67.2$$

SECTION – B

81. 00004.00

Sol. Given limit has value 4.

82. 00006.00

$$\text{Sol. } |S_1 P - S_2 P| = 2a$$

$$= 2 \times 3$$

$$= 6$$

83. 00002.00

$$\text{Sol. Required distance is } \frac{3\sqrt{2}}{4}, \text{ therefore value of } k \text{ is } 2.$$

84. 00002.00

$$\text{Sol. Required distance is } \sqrt{\frac{2}{3}}, \text{ therefore required value of } k \text{ is } 2.$$

85. 00004.00

$$\text{Sol. } \tan^4 x - 2 \tan^3 x - \tan^2 x + 2 \tan x + 1$$

$$\tan^4 x + \tan^2 x - 2 \tan^3 x + 2 \tan x - 2 \tan^2 x + 1$$

$$= (\tan^2 - \tan x)^2 + 2(\tan x - \tan^2 x) + 1 = 4$$

86. 00000.60

$$\text{Sol. } 10b^2 = 2560 \Rightarrow b = 16$$

$$\Rightarrow 1 - e^2 = \frac{16}{25} \Rightarrow e = \frac{3}{5}$$

87. 00090.00

Sol. For $c = 2b - a$, given line always passes through focal chord of parabola. Therefore required angle is 90° .

88. 02021.00

Sol. $(a^2 + 4a + 3)x^2 + (a^2 - a - 2)x + (a + 1)a = 0$
 $\Rightarrow (a + 1)(a + 3)x^2 + (a + 1)(a - 2)x + (a + 1)a = 0$
 $(a + 1)[(a + 3)x^2 + (a - 2)x + a] = 0$
 If $a = -1$, then quadratic have more than two roots.

89. 00015.00

Sol. Equation of the normal to the curve $x^2 = 4y$ is $x = my - 2m - m^3$, passes through the point (1, 2)
 $\Rightarrow m = -1$.

90. 00018.00

Sol. Given hyperbola is $3x^2 - 2y^2 = 6$ i.e., $\frac{x^2}{2} - \frac{y^2}{3} = 1$
 Let tangents from the point (α, β) be $y = mx \pm \sqrt{a^2m^2 - b^2}$ i.e. $(y - mx)^2 = a^2m^2 - b^2$
 $\Rightarrow (\beta - m\alpha)^2 = 2m^2 - 3 \quad (\because a^2 = 2 \text{ and } b^2 = 3)$
 $\Rightarrow m^2(\alpha^2 - 2) - 2\alpha\beta m + \beta^2 + 3 = 0$; let m_1, m_2 be the roots of the equation
 $\Rightarrow m_1m_2 = \frac{\beta^2 + 3}{\alpha^2 - 2} = \tan\theta \cdot \tan\phi = 2 \quad (\because \tan\theta \cdot \tan\phi = 2 \text{ given})$
 $\therefore \beta^2 + 3 = 2(\alpha^2 - 2)$
 $\Rightarrow \beta^2 = 2\alpha^2 - 7$
 $\Rightarrow 2\alpha^2 - \beta^2 = 7$